

MATH 4720 / MSSC 5720 **Sec 104** - Statistical Methods

Fall 2026

Instructor: Dr. Cheng-Han Yu

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Office Hours: TuTh 4:50–5:50 PM, Wed 12-1 PM, Cudahy Hall 353

Help Desk Hours: To be announced.

Website: math4720-f26.github.io/website/

Class Hours: TuTh 3:30-4:45 PM

Class Room: Cudahy Hall 118

1 Course Objectives

This course introduces the ideas of data summary and visualization, probability, and statistics, with applications to the natural and social sciences and daily life. Topics include random variables, discrete and continuous probability distributions, sampling distributions, confidence intervals, hypothesis testing, analysis of variance, and linear regression. This course builds a foundation for statistical inference, machine learning, and data modeling.

2 Prerequisites

MATH 1400, 1410, or 1450. Basic computer and internet use is expected. The course will also assume familiarity with using the internet and a personal computer. A portion of the course involves programming in [R](#) using [RStudio](#), but prior programming experience is not required.

3 Textbooks

- [IS] (optional) [Introduction to Statistics](#), by Dr. Cheng-Han Yu. (my online book)
- [IMS] (optional) [Introduction to Modern Statistics, 2nd edition](#), by Mine Çetinkaya-Rundel and Johanna Hardin. Publisher: OpenIntro. (computation- and data-oriented)

4 Course Management

- All course materials are posted on our course website <https://math4720-f26.github.io/website/>.
- Course grades are saved and managed in [D2L](#) > **Assessments** > **Grades**.
- Coursework is submitted to [D2L](#) > **Assessments** > **Dropbox**.
- Check [D2L](#) > **News** for new announcements.

5 Email Policy

- I will attempt to reply to your email quickly, usually **within 24 hours**.
- **Expect a reply on Monday if you send a question over the weekend.** If you do not receive a response from me within two days, resend your question/comment in case there was a “mix-up” with email communication (I hope this will not happen!).
- Please start your email subject line with **[math4720]** or **[mssc5720]** followed by a clear description of your question. See an example in Figure 1.

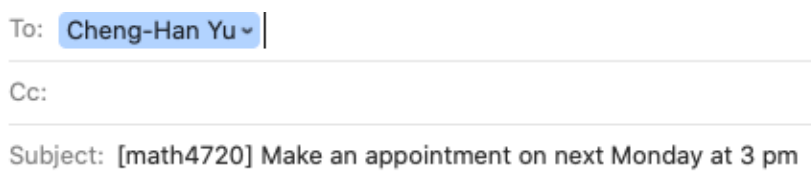


Figure 1: Email Subject Line Example

- Email etiquette is important. Please read this [article](#) to learn more about email etiquette.
- I am more than happy to answer your questions about this course or statistics in general. However, due to time constraints, I may choose **NOT** to respond to students’ emails if
 1. The student could answer their own inquiry by reading the syllabus or information on the course website or D2L.
 2. The student is asking for an extra credit opportunity. The answer is “no”.
 3. The student is requesting an extension on homework. The answer is “no”.
 4. The student is asking for a grade to be raised for no legitimate reason. The answer is “no”.
 5. The student is sending an email without proper etiquette.

6 Grading Policy

Category	Weight
Homework	10%
Quiz	20%
Exam	50%
Worksheet	10%
Group Activity	10%

- You will **NOT** be allowed to complete any extra-credit projects, homework, or exams to compensate for a poor average. Everyone must be given the same opportunity to do well in this class. Individual exams will **NOT** be curved.
- The final grade is based on your earned percentage and the grade-percentage conversion table. $[x, y)$ means greater than or equal to x and less than y . For example, 94.1 is in $[94, 100]$, so the grade is A, and 92.8 is in $[90, 94)$, so the grade is A-.

Table 2: Grade-Percentage Conversion

Grade	Percentage
A	[94, 100]
A-	[90, 94)
B+	[87, 90)
B	[83, 87)
B-	[80, 83)
C+	[77, 80)
C	[73, 77)
C-	[70, 73)
D+	[65, 70)
D	[60, 65)
F	[0, 60)

- This is not a course in which most students receive a grade of A. If you want to earn a good grade, study hard. No pain, no gain.

7 Homework

- Homework will be assigned through the course website.
- To submit your homework, please go to **D2L > Assessments > Dropbox** and upload your homework in **PDF** format.
- There will be 8 homework sets.
- Each homework assignment is due by **Friday at 11:59 PM (Do not miss it. This is a hard deadline. No late submissions will be accepted)**.
- MSSC 5720 students may have more or different homework questions.
- Homework is graded as *complete* or *incomplete*. Each incomplete set results in a 2% deduction from your overall grade.
- Generative AI (GenAI) is allowed, but you **must carefully cite it or reveal your use of AI**. See Section [12](#) for more details.

8 Quizzes

- There will be four in-class, 15-minute quizzes on Thursdays.
- Quizzes are individual and are in **closed-book** and **no-tech** format, except for a calculator.
- No cheat sheets or GenAI are allowed.
- **No make-up quizzes** will be given unless you have an [excused absence](#).
- Each quiz is worth 5% of your overall grade.
- If you miss a quiz due to an excused absence as defined in [Attendance in Academic Regulations](#), the 5% allocated to that quiz will be transferred to either the midterm exam or the final exam, whichever is

closest in time and has not yet been graded. For example, if you miss Quiz 2 and the midterm exam has not yet taken place, then the midterm exam will be worth 30% instead of 25%.

- If you miss more than one quiz, only one missed quiz can be replaced by transferring its weight to an exam. Any additional missed quizzes will receive a score of 0%.

9 Exams

- There will be two exams: the midterm exam and the final exam.
- Each exam has an in-class part and a take-home part. The in-class part tests your understanding of mathematical and statistical intuitions. The take-home part tests your ability to conduct statistical data analysis using statistical software such as R.
- For the in-class part, one **letter-size** cheat sheet is allowed. It must be turned in with your exam.
- For the take-home part, you are allowed to use GenAI tools. However, you **must carefully cite it or reveal your use of AI**. See Section 12 for more details.
- Please go to **Assessments > Dropbox** to submit your take-home exam in **PDF** format.
- The midterm exam covers **Weeks 1 to 7**. The final exam covers **all course material**.
- **No make-up exams** will be given for any reason unless you have an [excused absence](#).
- If you miss the midterm exam due to an excused absence defined in [Attendance in Academic Regulations](#), its 25% weight will be added to your final exam, i.e., $25\% + 25\% = 50\%$.

10 In-Class Worksheets

- Several worksheets will be distributed and completed during class sessions.
- The worksheets are graded as *complete* or *incomplete*.
- Each incomplete sheet results in a 1% deduction from your overall grade.

11 In-Class Group Activities

- There are several in-class group activities.
- The group activities are graded as *complete* or *incomplete*.
- Each incomplete activity results in a 2% deduction from your overall grade.

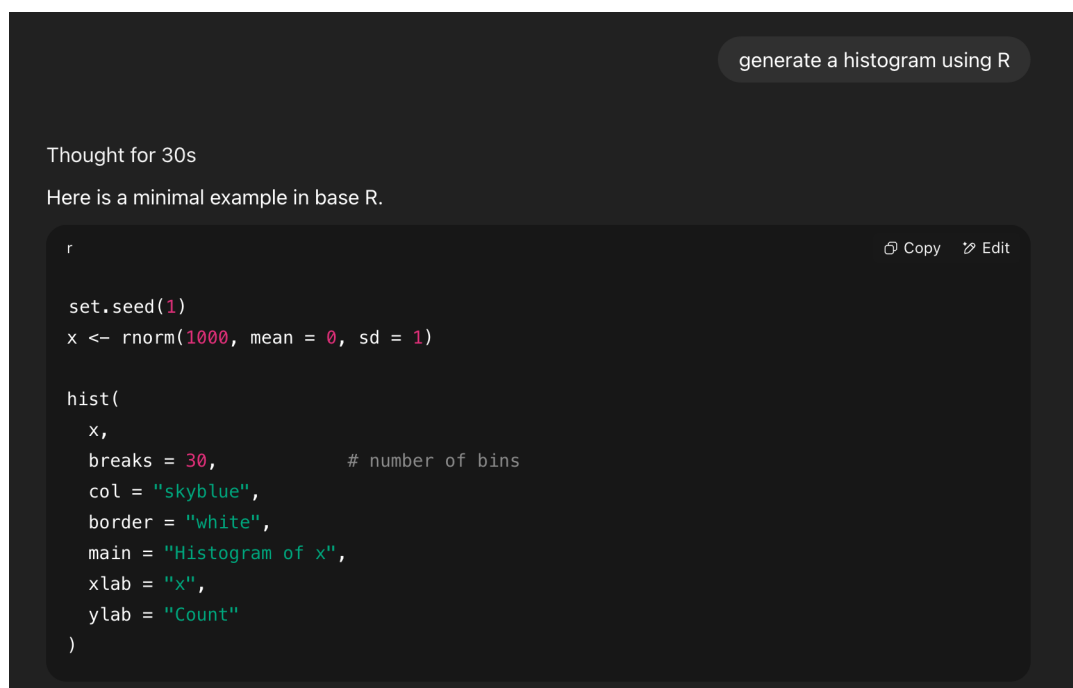
12 Generative Artificial Intelligence (GenAI) Policy

- *You are responsible for the content of all work submitted for this course.*
- For your homework, take-home exams, or any take-home work, you are allowed to use generative AI tools such as ChatGPT to generate a draft of your work.
- To avoid academic integrity issues, you **must cite your use of AI or screenshot your entire AI usage history**. Check the following resources for guidance on how to cite it.

- [How to cite ChatGPT](#)
- [How to Cite AI-Generated Content](#)
- If you use GenAI, please include the following in your submitted work:
 - **How I used AI (prompts or questions)**
 - **Generated output (screenshot or copy-paste excerpt)**
 - **How I used the output**

Here is an example.

- **How I used AI (prompts and questions)**
 - *I asked ChatGPT to generate a histogram using R.*
- **Generated output (screenshot or copy-paste excerpt)**



The screenshot shows a ChatGPT interface with a dark theme. At the top right, there is a button that says "generate a histogram using R". Below this, the text "Thought for 30s" is displayed. The main response from the AI is "Here is a minimal example in base R." followed by a code block containing R code for generating a histogram. The code is as follows:

```
r
set.seed(1)
x <- rnorm(1000, mean = 0, sd = 1)

hist(
  x,
  breaks = 30,          # number of bins
  col = "skyblue",
  border = "white",
  main = "Histogram of x",
  xlab = "x",
  ylab = "Count"
)
```

At the top right of the code block, there are icons for "Copy" and "Edit".

- **How I used the output**
 - *I reviewed the suggestions, but I did not use the exact code. Instead, I changed the code format and set the breaks value to 50.*

13 Academic Integrity

- Watch this [video](#) about academic integrity issues related to using GenAI.
- This course expects all students to follow University and College statements on [academic integrity](#).
- **Honor Pledge and Honor Code:** *I recognize the importance of personal integrity in all aspects of life and work. I commit myself to truthfulness, honor, and responsibility, by which I earn the respect of others. I support the development of good character, and commit myself to uphold the highest standards of academic integrity as an important aspect of personal integrity. My commitment obliges me to conduct myself according to the Marquette University Honor Code.*

14 Accommodation

If you need to request accommodations, or modify existing accommodations that address disability-related needs, please contact [Disability Services](#).

15 Tentative Course Schedule

Week 1, 8/31 - 9/6: Statistics and Data Science; Statistical Studies and Data Collection; Data Types

- Syllabus; IS Ch 1 - 2
- HW 1 issued

Week 2, 9/7 - 9/13: R Software and Programming

- IS Ch 3
- [Labor Day \(9/7\)](#)
- [Drop deadline by 11:59 PM on Tuesday, 9/8](#)
- [HW 1 due by 11:59 PM on Friday, 9/11](#)

Week 3, 9/14 - 9/20: Summarizing Data

- IS Ch 5 - 6
- HW 2 issued

Week 4, 9/21 - 9/27: Probability Fundamentals and Random Variables

- IS Ch 7 - 9
- HW 3 issued
- [HW 2 due by 11:59 PM on Friday, 9/25](#)
- [Quiz 1](#)

Week 5, 9/28 - 10/4: Discrete and Continuous Probability Distributions

- IS Ch 10 - 11
- [HW 3 due by 11:59 PM on Friday, 10/2](#)

Week 6, 10/5 - 10/11: Sampling Distributions and Central Limit Theorem

- IS Ch 12 - 13
- HW 4 issued

Week 7, 10/12 - 10/18: Confidence Intervals

- IS Ch 14
- [Quiz 2](#)
- [HW 4 due by 11:59 PM on Friday, 10/16](#)

Week 8, 10/19 - 10/25: Midterm Exam

- IS Ch 16

- HW 5 issued
- **In-class Midterm Exam on Tuesday, 10/20**
- **No class on Thursday, 10/22 (Midterm Break)**
- **Take-home Midterm Exam due by 11:59 PM on Friday, 10/23**

Week 9, 10/26 - 11/1: Hypothesis Testing

- **Midterm grade submission by noon on Tuesday, 10/27**
- IS Ch 16
- **HW 5 due by 11:59 PM on Friday, 10/30**

Week 10, 11/2 - 11/8: Hypothesis Testing

- HW 6 issued
- **Quiz 3**

Week 11, 11/9 - 11/15: Comparing Two Population Means

- IS Ch 17
- **HW 6 due by 11:59 PM on Friday, 11/13**

Week 12, 11/16 - 11/22: Inference about Population Variances

- IS Ch 24
- **Withdrawal deadline on Friday, 11/20**

Week 13, 11/23 - 11/29: Analysis of Variance

- IS Ch 26
- HW 7 issued
- **No class on Thursday, 11/26 (Thanksgiving)**

Week 14, 11/30 - 12/6: Analysis of Variance; Correlation

- IS Ch 27
- HW 8 issued
- **HW 7 due by 11:59 PM on Friday, 12/4**

Week 15, 12/7 - 12/13: Linear Regression; Categorical Data Analysis

- IS Ch 19, 20, 27
- **Quiz 4**
- **HW 8 due by 11:59 PM on Friday, 12/11**

Week 16, 12/14 - 12/20: Final Exam

- **In-class Final Exam on Thursday, 12/17, from 8 to 10 AM**

Week 17, 12/21 - 12/27

- **Final grade submission by noon on Tuesday, 12/22**

Dr. Yu reserves the right to change the syllabus.